

1 (a)(i)	Period of Earth, $T = 24 \times 3600 \text{ s} (= 86\,400 \text{ s})$ Angular velocity $\omega = 2\pi/T$ $= 7.27 \times 10^{-5} \text{ rad s}^{-1}$ (shown)	[1]
(a)(ii)	$a = r\omega^2$ $= (6.38 \times 10^6) \times [\text{Ans in (b) (i)}]^2$ $= 0.0337 \text{ m s}^{-2}$	[1] [1- ecf]
(b)(i)	- There is circular motion at the equator but not at the poles, - At the equator, part of the gravitational force is used to provide the centripetal force (required to keep any mass located there in circular motion, centripetal force is provided by the gravitational force of Earth) (Hence only the remaining part of the gravitational force is available to provide the acceleration of free fall.) Or - Gravi force of Earth $= (mg + ma_c)_{\text{equa}} = (mg + ma_c)_{\text{pole}}$. [1] - Since $a_c \text{ at pole} = 0$, $g_{\text{pole}} > g_{\text{equa}}$, ie g is different for the 2 locations [1] Note: g = acceleration of free fall as defined by the question not gravitational field strength. Be careful in use of symbols.	[1] [1]
(b)(ii)	- Value of g, at the pole is 9.81 m s^{-2} - Since centripetal acceleration at equator (in b(ii)) is 0.0337 m s^{-2}, compared to 9.81 m s^{-2}, the difference in g at these 2 locations is small.	[1] [1]
(c)(i)	The gravitational force of attraction between two point masses is proportional to the product of their masses and inversely proportional to the square of their separation.	[1]
(c) (ii)	(By Newton's law), Gravitational force on a (point) mass m at Earth's surface, $F = \frac{GMm}{R^2}$ Since field strength g is the gravitational force per unit mass at that point, $g = F/m = \frac{GM}{R^2}$	[1] [1]
(d)(i)	{ Since the gravitational force provides the centripetal force, } $\frac{GMm}{r^2} = m\omega^2 r \quad \{\text{where } r = \text{dist of satellite fr Earth's centre}\}$	[1]

	$(\square r^3 = GM/\omega^2, \text{ not assessed}) \quad \text{-- (1)}$ • From (c) , $GM = gR^2 \quad \text{-- (2) } \{ R = \text{Earth's radius} \}$ (2) into (1): $(r^3 = gR^2/\omega^2, \text{ not assessed})$ • Substituting, $r = 4.23 \times 10^7 \text{ m}$ $\{g = 9.81 \text{ m s}^{-2}, R = 6.38 \times 10^6 \text{ m}, \omega = 7.27 \times 10^{-5} \text{ rad s}^{-1}\}$	[1] [1]
(d)(ii)	1. Satellite lies on the equatorial plane of the earth 2. Satellite rotates/revolves from west to east or, following the Earth's rotation	[1] [1]

2(a)	Volume decreases : hence w is positive	[1]
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